SOEN 343 - Phase 1

SUMMS – Smart Urban Mobility Management System

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(a) Project definition:

Objectives

- Create a single platform that allows users to plan, reserve, and pay for multimodal trips (bike, scooter, shared car, parking, and public transit).
- Reduce urban congestion by recommending multimodal routes based on real-time conditions (traffic, availability).
- Reduce time spent searching for parking or available vehicles by showing live availability and allowing advance reservations.
- Offer real-time, personalized mobility suggestions to users that adapt to user preferences (e.g., bike vs scooter) and current city conditions (traffic, weather, demand).
- Provide city officials with live and historical analytics for planning and policy decisions (cross-city and cross-mode comparisons of usage, congestion, and emissions).
- Encourage sustainability among users by recommending lower emissions multimodal routes.
- Encourage users to responsibly use shared rental services using gamification (incentives and behavior-based analytics).

Users

SUMMS serves a diverse set of users, each interacting with the system in different ways:
- Citizens and commuters who use shared vehicles, parking facilities, and public transport.
- Mobility service providers, such as bike-sharing, car-sharing, and scooter operators.
- Public transportation operators managing buses, metros, or trams.
- City administrators responsible for urban mobility planning and regulation.
- System administrators who manage and maintain the platform.

Stakeholders

Key stakeholders of SUMMS include:

- Municipal governments and urban planning authorities.
- Public transportation agencies.
- Private shared mobility providers.
- Parking infrastructure operators.
- Environmental and sustainability organizations.
- Citizens and local communities.

Methodology and member responsibilities

Team Leader

- Coordinates tasks and deadlines
- Integrates all sections and handles submission
- Validates project artifacts (objectives, problem definition, diagrams) against requirements and rubric

Architect 1

- Leads system vision, validates objectives and problem definition
- Oversees context diagram consistency

Architect 2:

- Leads domain modeling and conceptual design
- Ensures alignment between models and requirements

Developer 1

- Contributes to problem definition and usage scenarios
- Assists with analytics and feature ideas

Developer 2

- Contributes to project definition and solution advantages
- Supports diagram refinement and validation

Tester & Documentation

- Selects collaboration and design tools and justifies them
- Manages shared repository (GitHub/Drive) and version control
- Handles formatting, figures, and final document quality

*<u>*Collaboration note</u>: All team members participated in brainstorming and peer reviews were done on each other's work. Diagrams and written sections (project goals and definition) were reviewed together to ensure consistency and quality, with weekly lab meetings to progress.*

Elements and deliverables of the project

Sprint 1 deliverables (scope, boundaries):

- Project definition
    - Clear objectives
    - Identified users and stakeholders
    - Defined project scope and boundaries

- Problem definition
    - Analysis of fragmented mobility systems
    - Identified gaps between current and desired urban mobility solutions
    - Justified SUMMS solution and its advantages
    - Defined incentive mechanisms and analytics for rental mobility

- Context diagram
    - All external entities (users, providers, mobility systems, payment systems)
    - System boundary and data flows

- Domain model (concepts and relationships)
    - Main concepts and their attributes (User, Trip, Vehicle, Reservation, ParkingSpot, City, etc.)
    - Relationships between concepts (mobility services, rentals, and analytics)

- Tool section and justification

(b) Problem definition:

Problem statement and emergence.

1. What's the problem?
Urban mobility services are fragmented. Users have to use separate apps for bike, scooter or car sharing, parking, and public transit. This causes inefficient trip planning, increased congestion, wasted time searching for vehicles or parking, and limited data for decision making for cities.

2. How did the problem emerge?

Cities added new mobility services over time: first public transit, then parking systems, then private bike, scooter or car sharing. Each system was built independently with its own app and data silo. No central platform exists to coordinate them in real time.

Proposed solution and its advantages.

3. What is the solution?

Smart Urban Mobility Management System (SUMMS): a centralized mobility management platform that:

- Combines real-time data from all mobility providers
- Lets users plan, reserve, and pay for multimodal trips in one place
- Optimizes routing using live availability, traffic, and transit data
- Provides dashboards for city administrators and providers

4. What are the advantages of SUMMS compared to existing solutions?

- One account instead of many apps
- Real-time optimization across different mobility services (e.g., scooter → metro → parking)
- City-wide analytics across all mobility types, not just one provider
- Personalized recommendations based on user preferences and habits
- Neutral city-level platform, not owned by a single private operator

5. What features can make SUMMS better than competitors?

- Gamified rental reliability system
  Users earn reliability scores based on behaviors such as on-time returns and correct parking. Higher scores unlock benefits like priority reservations or discounts.

- Behavior-driven incentive engine
  Users are rewarded for actions that improve system efficiency, such as returning vehicles to high-demand areas or avoiding peak congestion.

- Trip-level sustainability analytics (CO_2 impact)
  Each rental trip includes an estimated environmental impact score based on distance and vehicle type.

- Predictive rental availability analytics (predict bikes/scooters/parking availability)
  Historical and real-time data are used to forecast short-term availability of bikes, scooters, and shared cars.

- City-level analytics dashboards

Administrators can analyze demand patterns, fleet usage, and user behavior to support planning and optimization.

(c) Technology used: Tools for collaboration, monitoring, design, and coding.

Collaboration and Monitoring

- **Google Docs**: shared writing and storage
- **GitHub**: version control and manage repository, sprints, milestones, and tasks.
- **WhatsApp**: team communication

Design and Coding Tools

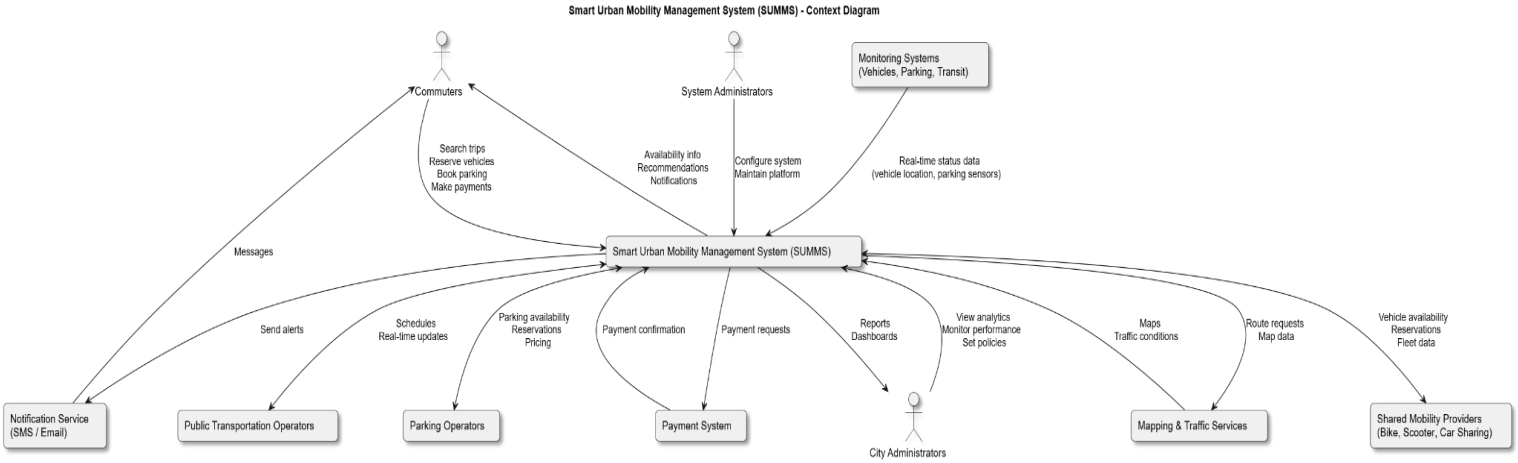
- **Draw.io**: context and domain diagrams
- **VS Code**: documentation and future coding

Programming Languages:

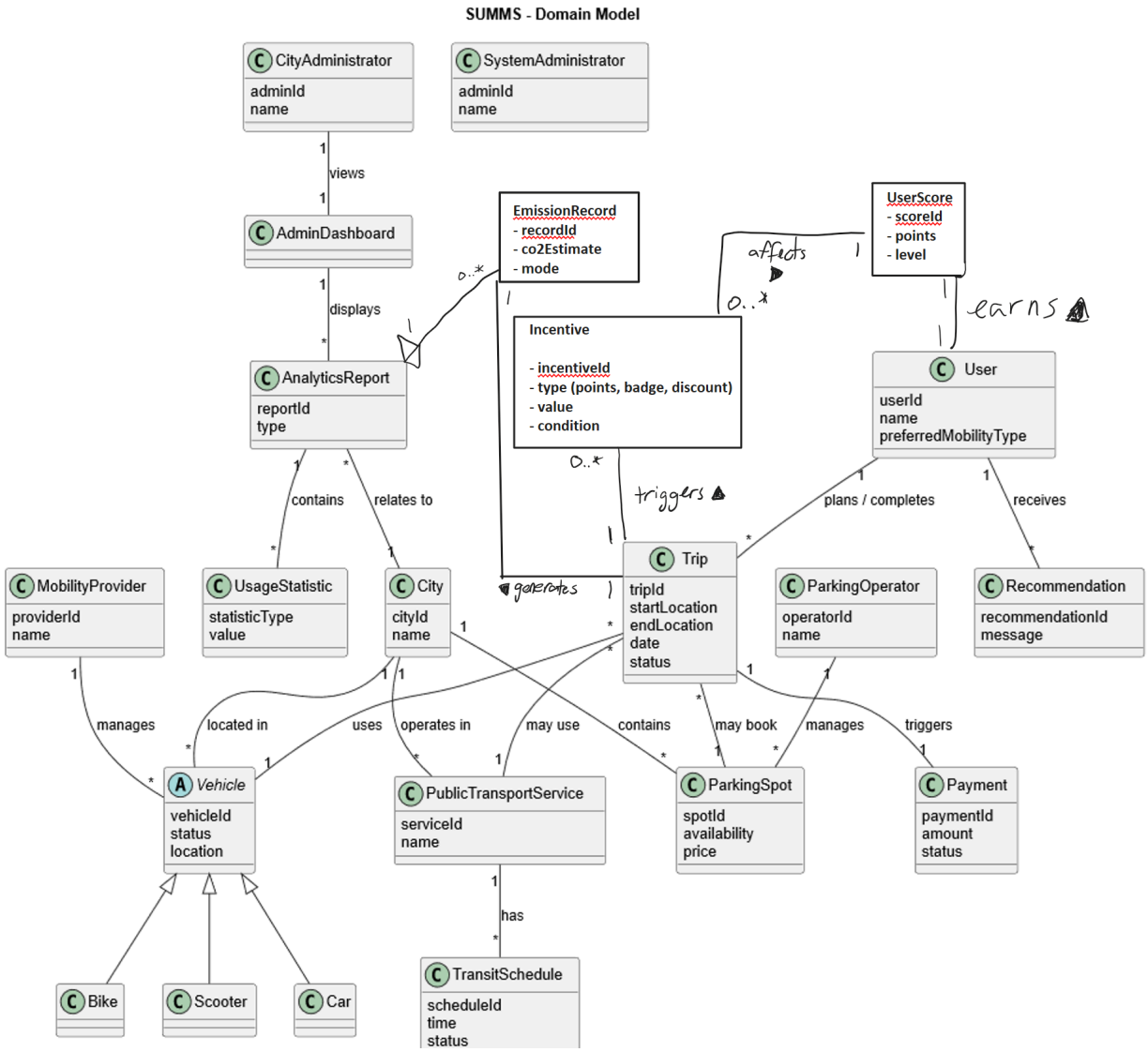
- **TypeScript (React)**: user and admin interfaces
- **Java**: backend services and core system logic
- **PostgreSQL**: structured data storage
- **Python**: analytics and sustainability calculations
- **JSON / REST**: communication with external systems

These tools allow us to collaborate efficiently as a team and implement any changes to diagrams, documents, or other artifacts. The tech stack chosen is what most team members are familiar with.

(d) Context diagram



(e) Domain model



(f) Conclusion: Brief summary of the phase 1 of the project.

Phase 1 defines the problem of fragmented urban mobility services and proposes SUMMS as a unified city-level platform. The context diagram and domain model identify the main actors, services, and interactions which will help later design and implementation phases. The solution emphasizes rental system optimization and analytics-driven features, with a focus on sustainability, efficiency, and responsible user behavior.